

Analysis of Single Pauli Fault Propagation in Steane Code Syndrome Extraction Using Flag Qubits

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Abstract

Fault-tolerant syndrome extraction is essential to making quantum error correction viable, since a single physical fault can propagate uncontrollably through the syndrome measurement circuit and produce an error too large for the underlying code to correct. This project compares two syndrome-extraction techniques for a representative X-type stabilizer of the $[[7,1,3]]$ Steane code. It utilizes a conventional five-qubit ancilla-based circuit and a six-qubit flag-qubit-based circuit that includes an auxiliary flag ancilla to catch dangerous fault propagation. Every possible single-qubit Pauli fault (X, Y, or Z) was injected at each non-measurement gate location, and its effect was followed using a Heisenberg-picture propagation model, then independently verified by simulating the faulty circuit on a state-vector simulator. Faults were then classified as correctable, detected by the flag, or not correctable based on the weight of the propagated data error and whether the fault also flipped the flag qubit's outcome. The flag-qubit circuit reduced the rate of undetected, uncorrectable errors from 20% in the baseline circuit to 4.8%. This reduction of over 75% was only at the cost of one additional qubit and three additional gates. The results demonstrate the importance of fault-tolerant gadgets, particularly for determining how effectively a syndrome-extraction circuit mitigates fault propagation.

keywords: Quantum Error Correction, Fault Tolerance, Steane Code, Flag Qubits.

1. INTRODUCTION

Quantum computers have the potential to solve problems that are impractical for classical computers, but they are also highly susceptible to errors caused by environmental noise and imperfect quantum operations. Even a single fault can alter the state of a qubit, and if that fault spreads through a quantum circuit, it can become too large for an error-correcting code to recover. As quantum devices continue to scale, developing reliable methods for detecting and containing these errors has become one of the most important challenges in quantum computing.

Quantum error correction provides a way to protect logical quantum information by encoding it across multiple physical qubits. However, successful error correction depends not only on the code itself but also on how error syndromes are measured. Faults occurring during syndrome extraction, particularly on an ancilla qubit, can propagate through controlled operations

and create correlated errors on multiple data qubits. If these propagated errors exceed the correction capability of the code, they may result in logical failure. Fault-tolerant syndrome-extraction circuits are designed to prevent this by ensuring that a single physical fault remains localized and does not spread into an uncorrectable error.

This project investigates fault propagation in the $[[7,1,3]]$ Steane code by comparing two syndrome-extraction approaches: a conventional ancilla-based stabilizer measurement circuit and a flag-qubit stabilizer measurement circuit. Single-qubit Pauli faults (X, Y, or Z) are injected at every viable location in each circuit to examine how errors propagate, which faults are detected, whether the resulting data errors remain correctable, and how the two approaches compare in terms of qubit and gate overhead. By reconstructing and testing both circuits using quantum circuit simulation, this project provides a transparent comparison of their fault-tolerant

behavior and demonstrates how circuit design influences the reliability of quantum error correction.

2. LITERATURE REVIEW

Quantum computing is the processing of information according to the fundamental laws of physics [1]. This technique strives to outperform traditional classical computing methods in efficiency and accuracy. However, quantum information is extremely fragile and disrupted by even the smallest amounts of environmental noise and inaccuracies [2]. These disturbances introduce errors such as bit flips, which change a qubit's state, and phase flips, which alter the relative phase of a quantum state [2]. Protecting quantum information requires specialized quantum error correction techniques rather than conventional error correction methods. Quantum error correction (QEC) addresses this challenge by predicting error-correcting "codeword" quantum states, which unfortunately require many physical qubits [1].

Derived from a classical [7,4,3] Hamming code, the $[[7,1,3]]$ Steane code is a well-known QEC code that encodes a single logical qubit into seven physical qubits with distance three and corrects any arbitrary physical error [3]. As a CSS stabilizer code, it separates X- and Z-syndrome measurements and uses independent classical codes to correct bit-flip and phase-flip errors while preserving the encoded logical state. For each stabilizer, one ancilla qubit is prepared, entangled with the data qubits, and measured to determine the error syndrome before the appropriate Pauli correction is applied [3].

The main advantage of the Steane code is the independence between bit-flip and phase-flip errors. This is achieved through transversal implementation of Clifford gates, where corresponding physical qubits interact only with their counterparts in another code block, reducing error propagation during computation [3].

Recently proposed fault-tolerant parity-check measurements have reduced overhead in both qubits and gate operations. Traditional

syndrome-extraction methods may require several auxiliary qubits, whereas flag-based approaches achieve fault tolerance using only a syndrome qubit and one additional flag qubit. Hilder et al. (2022) demonstrated this approach using a weight-four parity-check measurement on a trapped-ion quantum computer. They showed that faults on the syndrome ancilla can propagate into correlated data-qubit errors and that the flag qubit reliably detects these dangerous propagation events, improving parity-measurement fidelity from 92.3% to 93.2% [4]. Similarly, Reichardt (2020) showed that multiple syndromes can be measured in parallel using flag qubits, reducing both ancilla overhead and sensitivity to memory noise [5].

Collectively, these studies establish the $[[7,1,3]]$ Steane code as a foundation for fault-tolerant quantum error correction while demonstrating how different syndrome-extraction strategies influence fault propagation. Although flag-qubit approaches have been shown to detect correlated errors while reducing resource overhead, these circuits are often introduced in isolation and evaluated under different assumptions, making direct comparison difficult. This project addresses that gap by reconstructing and testing conventional ancilla-based and flag-qubit stabilizer-measurement circuits for the $[[7,1,3]]$ Steane code. By injecting representative single-qubit Pauli faults into both circuits, the study examines how faults propagate, which faults are detected, whether the resulting data errors remain correctable, and how the two approaches differ in qubit and gate overhead.

3. METHODOLOGY

3.1. Steane Code Stabilizer

The $[[7,1,3]]$ Steane code encodes one logical qubit into seven physical data qubits and has three independent X-type and three independent Z-type stabilizer generators, each of weight four [3]. This project focuses on a single representative stabilizer generator, the X-type stabilizer supported on the data qubits 3, 4, 5, and 6, so that fault behavior can be studied in detail rather than an analysis of the average across the full stabilizer group. Two additional qubits are

used to extract this stabilizer: a syndrome ancilla, and, in the flag-qubit circuit, a flag ancilla.

Before injecting any faults, the circuits were validated by preparing the data qubits in a fixed test state consistent with a +1 eigenvalue of the stabilizer being measured. This clean-state preparation was built from Hadamard gates and a fixed sequence of CNOTs, and was used strictly as a sanity check. Running the baseline circuit against this clean state should always yield a syndrome measurement of 0, confirming that we had built the stabilizer circuit correctly before injecting any faults.

3.2. Baseline Syndrome-Extraction Circuit

The baseline circuit follows the standard construction for measuring an X-type stabilizer with a single ancilla, shown in Figure 1. The syndrome ancilla is prepared in the $|+\rangle$ state with a Hadamard gate, then used as the control qubit of a CNOT gate targeting each of the four data qubits in the stabilizer's support, in sequence. The ancilla is returned to the computational basis with a second Hadamard gate and measured; the measurement outcome gives the stabilizer's syndrome bit. This circuit uses five qubits in total (four data qubits plus the ancilla) and seven gates (two Hadamards and four CNOTs, plus the measurement).

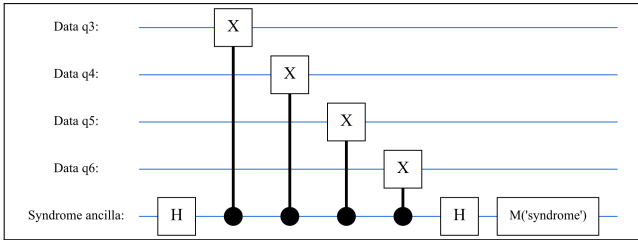


Figure 1. Circuit diagram of the baseline ancilla-based circuit

3.3. Flag-Qubit Syndrome-Extraction Circuit

The flag-qubit circuit measures the same stabilizer using the same syndrome ancilla and the same four data-qubit targets, but inserts a flag ancilla into the gate sequence to catch a specific dangerous failure mode, as shown in Figure 2. Here, a fault on the syndrome ancilla partway through its CNOTs to the data qubits, and that then propagates to more than one data qubit before the ancilla is measured.

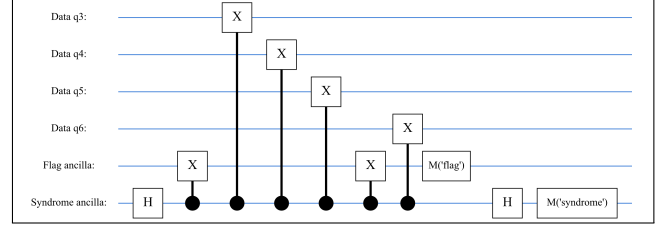


Figure 2. Circuit diagram of the flag-qubit based circuit

The flag qubit starts in the state $|0\rangle$ and is coupled to the syndrome ancilla with a CNOT immediately after the ancilla is prepared in $|+\rangle$, opening a "flag window." The first three of the four data-qubit CNOTs are performed while this window is open. A second CNOT between the syndrome ancilla and the flag qubit then closes the window, after which the final data-qubit CNOT is applied. The syndrome ancilla is returned to the computational basis with a Hadamard gate, and both the syndrome ancilla and the flag qubit are measured. A fault that occurs on the syndrome ancilla while the flag window is open should also flip the flag qubit's measurement outcome, signaling that the resulting data error should not be treated as an ordinary, independent single-qubit error. This circuit uses six qubits (the four data qubits, the syndrome ancilla, and the flag qubit) and ten gates (two Hadamards, six CNOTs, and two measurements).

3.4. Fault Injection and Propagation Model

Fault tolerance was tested using a single-fault model: for every location in a circuit where a qubit is acted on by a non-measurement gate, each of the three single-qubit Pauli faults (X, Y, or Z) was considered separately, one at a time. No two faults were injected in the same run, to analyze how one physical fault behaves rather than on multi-fault or correlated noise processes.

For each location and Pauli fault pair, the fault's effect was tracked forward through the rest of the circuit using the standard Heisenberg picture for Clifford circuits, in which each qubit's error is represented as a pair of bits indicating whether an X and/or a Z component is present. A Hadamard gate exchanges these two components, a CNOT gate propagates the X component from its control qubit to its target, and propagates the Z component from its target back to its control, and a Pauli gate commutes with itself and leaves

the error frame unchanged, which is what allows the fault itself to be inserted as an ordinary Pauli gate without disturbing this model. Propagating the fault in this way through every remaining gate in the circuit gives the final Pauli error, if any, on every qubit, including the data qubits, the syndrome ancilla, and the flag qubit where present.

In parallel with the propagation, the fault was also injected directly into a copy of the circuit as an explicit Pauli gate, and the resulting circuit was run on a full state-vector quantum circuit simulator to obtain an actual measured syndrome outcome. This was for a second, independent check that the propagation model’s predictions were consistent with the behavior of the simulated circuit itself.

3.5. Fault Classification

Each injected fault was classified into exactly one of three outcomes, based on its propagated effect on the data qubits and, where a flag qubit is present, on whether that fault also flipped the flag qubit’s measurement outcome:

- **Detected by flag:** The fault does not flip the flag qubit (or the circuit has no flag qubit, as in the baseline), and the propagated error affects at most one data qubit, which is within the single-error correction capability of the Steane code. fault flips the flag qubit’s outcome. Because the flag circuit has no way to inspect the actual weight of the resulting data error at decode time, any fault that trips the flag is classified this way regardless of how many data qubits it ultimately affects, including cases where the propagated data error would otherwise have been trivially correctable.
- **Correctable:** The fault does not flip the flag qubit (or the circuit has no flag qubit, as in the baseline), and the propagated error affects at most one data qubit, which is within the single-error correction capability of the Steane code.
- **Not Correctable:** The fault does not flip the flag qubit, and the propagated error affects more than one data qubit, exceeding what the Steane code can correct on its own.

This same three-way classification was applied to every fault in both the baseline circuit (which has no flag qubit, and so is scored only on the correctable/not-correctable boundary) and the flag circuit, so that the two circuits’ fault-handling behavior could be compared directly and on equal terms.

3.6. Overhead Comparison

Finally, the baseline and flag circuits were compared on resource overhead, which was counted as the total number of qubits involved in each circuit and the total number of gate operations each circuit requires. This comparison isolates the physical cost of adding the flag mechanism (one additional ancilla qubit and the extra gates and measurement needed to operate it) separately from its benefit in catching otherwise-uncorrectable faults, which is reported alongside the fault-classification results.

4. RESULTS AND DISCUSSION

Fault Outcome Comparison

The baseline circuit has 10 fault locations, giving 30 injected faults (10 locations $\times$ 3 Pauli types); the flag circuit has 14 fault locations, giving 42 injected faults. The classification of every injected fault is summarized in Table 1 and shown graphically in Figure 3.

Table 1. Fault-outcome breakdown for the baseline and flag circuits.

Outcome	Baseline Circuit	Flag Circuit
Correctable	80.0% (24/30)	66.7% (28/42)
Detected by Flag	-	28.6% (12/42)
Not Correctable	20.0% (6/30)	4.8% (2/42)

In the baseline circuit, every uncorrectable fault originates on the syndrome ancilla. Because the ancilla is used as the control qubit for all four data-qubit CNOTs, an X or Y fault on the ancilla propagates forward to every data qubit whose CNOT has not yet occurred. A fault injected immediately after ancilla preparation propagates to all four data qubits (weight 4), while one injected just after the last CNOT propagates to none (weight 0). On the other hand, a Z fault on the ancilla never propagates to the data qubits at all under this gate ordering, since a CNOT only carries the Z component from target back to

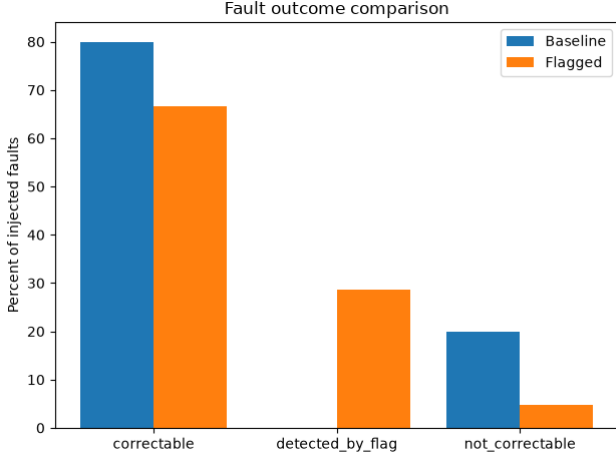


Figure 3. Grouped bar chart of the percentages in Table 1.

control, not the other way around. Faults injected directly on a data qubit are always weight 1, since no further gate acts on that qubit afterward, and are therefore always correctable. This produces the six uncorrectable outcomes in the baseline: X and Y ancilla faults injected before each of the first three data-qubit CNOTs, corresponding to propagated weights of 4, 3, and 2 data qubits.

The flag circuit removes four of these six dangerous cases. The measurement flips whenever an X or Y fault on the ancilla is present while the flag window is open, correctly labeling those faults as "detected by flag" rather than allowing them to just pass silently as multi-qubit data errors. It also flags every fault injected directly on the flag qubit itself, even though such faults never touch the data qubits (weight 0). The flag circuit cannot actually distinguish a fault on itself from a fault it has correctly caught, so it treats both cases the same way, and this project classifies a flagged fault as "detected" regardless of the data-qubit weight it turns out to correspond to.

Overhead Comparison

Table 2 summarizes the resource cost of the flag mechanism.

Table 2. Qubit and gate overhead for the two circuits

Resource	Baseline Circuit	Flag Circuit	Extra
Qubits	5	6	+1
Gates	7	10	+3

For a single additional ancilla qubit and three

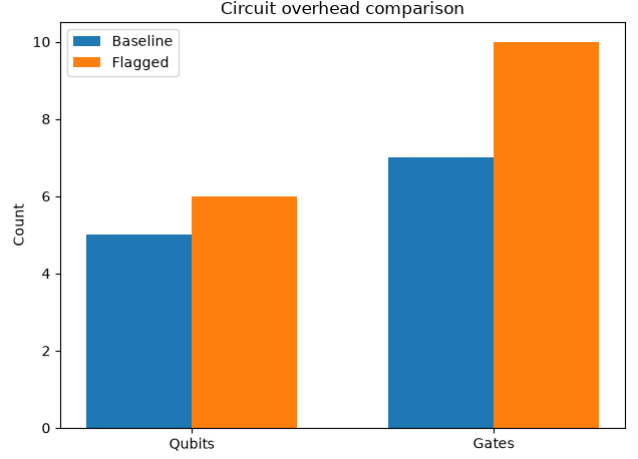


Figure 4. Grouped bar chart of the qubit and gate counts in Table 2.

additional gates (two CNOTs to couple the flag to the syndrome ancilla, plus one additional measurement), the flag circuit cuts the fraction of undetected, uncorrectable faults from 20.0% to 4.8%. This reduction is over 75% in the rate of dangerous, silent failures for this stabilizer.

Residual Uncorrectable Faults: The Gap Before the Flag Window Opens

The flag qubit does not close the gap entirely: 4.8% of faults in the flag circuit are still classified as not correctable. Tracing these two faults (an X and a Y fault, both injected on the syndrome ancilla immediately after ancilla preparation, before any other gate) reveals a specific limitation of this flag placement rather than a general failure of the flag-qubit approach.

An X or Y fault injected at this location has an X component that is still present on the ancilla when it reaches the CNOT that opens the flag window, so it flips the flag qubit's error frame at that point. However, the same persistent X component on the ancilla is still present, unchanged, when the ancilla reaches the second CNOT that closes the flag window later in the circuit. Because the ancilla's error value has not changed between the two flag CNOTs, the second CNOT flips the flag's error frame a second time, and the two flips cancel. The flag qubit's final measurement outcome is unaffected, even though the fault has already propagated to all four data qubits by the time the circuit finishes. In effect, a fault that occurs before the flag window opens, but persists on the same control line through both the opening

and closing of the window, is invisible to the flag by construction: it interacts with the flag qubit an even number of times.

This is a timing gap rather than a flaw in the flag mechanism itself: any fault injected after the flag window opens is caught, and every fault on the flag qubit itself is caught. The two residual failures both occur in the single moment between ancilla preparation and the window's opening CNOT. This suggests a straightforward refinement: moving the flag window's opening CNOT to immediately follow ancilla preparation, with no intervening moment, would remove this gap entirely for this stabilizer, since there would then be no fault location on the ancilla that precedes the window.

False Positives on the Flag Qubit Itself

Two of the twelve "detected by flag" outcomes never touch any data qubit; their propagated data-error weight is 0. The flag circuit still reports these as detected, since the classification treats any flag flip as signaling a potential correlated error, without attempting to separately verify whether a real data error actually occurred. This is actually consistent with how a flag qubit is meant to be used in practice: at the point the ancilla is measured, the circuit has no way to know the true weight of the propagated error, only whether the flag fired. Treating every flag flip as "detected" rather than trying to guess which flagged events were harmless is the conservative choice, at the cost of over-reporting a small number of harmless events as if they required special handling.

Summary of Results

Comparing the two circuits shows a clear trade-off between overhead and fault coverage. The flag circuit costs one extra qubit and three extra gates compared to the baseline, and in exchange reduces the rate of undetected, uncorrectable faults from 20.0% to 4.8% for this stabilizer. The residual uncorrectable faults are not evidence that flag qubits are ineffective in general; they trace to a specific, identifiable gap in this circuit's flag-window placement, and this gap appears fixable by tightening the window's placement rather than by adding further

hardware. This supports the broader claim in the fault-tolerance literature that the benefit of a flag qubit depends closely on exactly where in the gate sequence it is inserted, not just on whether one is present at all.

5. CONCLUSION

This project compared conventional ancilla-based and flag-qubit syndrome-extraction circuits for the $[[7,1,3]]$ Steane code by systematically injecting single-qubit Pauli faults and analyzing their propagation. The results showed that the flag-qubit circuit significantly improved fault tolerance, reducing undetected, uncorrectable faults from 20% to 4.8% while requiring only one additional qubit and three additional gates. Residual failures were traced to a timing gap in the flag-window placement, suggesting that future work should investigate alternative flag placements and gate orderings to further improve fault coverage.

Overall, this work demonstrates that effective quantum error correction depends not only on the underlying code but also on the design of the syndrome-extraction circuit. By reconstructing and comparing two syndrome-extraction approaches, this project provides a practical framework for studying fault propagation and evaluating future fault-tolerant circuit designs. Further work could extend this analysis to all Steane stabilizers, additional fault models, and larger-distance quantum error-correcting codes.

6. CODE AND DATA AVAILABILITY

The code for this project is available at FT-Gadget-Inspector. The circuit building, simulation, fault injection, and analysis were all done in Python.

7. ACKNOWLEDGMENTS

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